RAJA ABRAHEEM RASHID EJAZ

MSc Artificial Intelligence, Brunel University of London · Applicant for doctoral study in Artificial Intelligence
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RESEARCH PROFILE

I study how learning systems should behave when the world they run in does not hold still. My main line of work is **nested, multi-timescale learning**: letting different parts of a model update at different rates, which I have developed for federated learning, multi-agent reinforcement learning in 6G networks and adaptive-fidelity inference. A second line concerns **language-model agents**: when a reasoning model should stop thinking, how cheap guards can supervise scoped agents, and how agentic interfaces move risk rather than remove it. Across both I value results that survive honest testing, including negative ones: my MSc dissertation shows that no published early-exit halting signal generalises across domains, and explains why. I want to do open-ended, conceptually ambitious AI research in a collaborative group, drawing on applied work in wireless networks, healthcare, power systems and security.

Interests: multi-timescale and nested learning · LLM agents and reasoning · multi-agent reinforcement learning · federated and privacy-preserving learning · trustworthy, reproducible machine learning.

Record: 14 peer-reviewed papers published or accepted (2 journal, 12 conference; 8 as first author) · 10 under review · Best Paper Award, ICSL-DSGA 2026 · 3 co-supervised undergraduate projects produced accepted papers.

EDUCATION

MSc Artificial Intelligence, Brunel University of London 2024 – 2026
Distinction. Supervisor: Prof. Yongmin Li. *London, UK*

- **Dissertation:** *Knowing When to Stop Thinking: Risk-Controlled Early Exit for Reasoning Language Models.* Code: github.com/Abraheem13/TokenGuard.
- Compared five published halting signals and a confidence control on one shared probe stream across 12 benchmark and model settings (858 questions, 6,068 reasoning traces, 54,812 graded answers), judging every rule by its worst case on a held-out split.
- Showed that no fixed signal generalises (answer agreement gains 8.3 points on GSM8K with Qwen3-8B but loses 23.5 on AIME-2024 with Qwen3-4B) and traced the failure to *error stickiness*, which rises from 0.13 to 0.77 as the answer space shrinks; two tests designed after the model was formulated confirmed it.
- Proposed a Nested Token-Budget Controller whose fusion tier cuts worst-case regret to 0.031, against 0.100 for the best fixed signal, at 0.9 to 7.1% probing overhead.
- Modules: Advanced Deep Learning, Statistical Learning Theory, Convex Optimization, Information Theory.

BEng Electrical Engineering (AI specialisation), COMSATS University Islamabad 2020 – 2024
Islamabad, Pakistan. Final-year research project, funded by NESCOM: intelligent reconfigurable surfaces for UAV communication links.

PUBLICATIONS

Published as R. A. R. Ejaz or A. Rashid (own name in bold). † corresponding author. Status as of October 2026.

Journal articles

[J1] **A. Rashid**, F. Iradat, W. Iqbal, I. Syed, K. Khan, and K. Yahya. "Nested Multi-Agent Reinforcement Learning for Adaptive Resource Management in 6G Network Slicing: A Multi-Timescale Framework with Convergence Guarantees." *IEEE Open Journal of the Communications Society*, vol. 7, pp. 7624–7640, 2026. doi:10.1109/OJCOMS.2026.3708409

[J2] **R. A. R. Ejaz**, F. Iradat, W. Iqbal, and A. Ahmad. "Nested Learning with Attention-Guided Multi-Frequency Supervision for Hepatic Vessel Segmentation." *Neural Computing and Applications* (Springer). Accepted, 2026.

Conference papers (published or accepted)

[C1] **R. A. R. Ejaz**, Y. A. Bangash, F. Iradat, M. M. W. Iqbal, and S. V. Easwaramoorthy. "Hierarchical Bayesian Nested Optimisation for Uncertainty-Aware Resource Allocation in 6G Open-RAN." *Int. Conf. on Statistical Learning, Data Science and Generative AI (ICSL-DSGA 2026)*, Sunway University, Malaysia. **Best Paper Award.**

[C2] S. S. H. Hajjaj, **A. Rashid**, and S. V. Easwaramoorthy. "STREAM-HGAT: Mining Spatiotemporal Patterns and Anomalies in Smart-Grid Telemetry Streams with a Hybrid Graph-Attention Framework." *44th IEEE International Conference on Computer Design (ICCD 2026)*, Hong Kong.

[C3] S. V. Easwaramoorthy and **A. Rashid**. "HEAL-GTN: Mining Longitudinal Electronic Health Records with a Hierarchical Graph-Temporal Network for Early Sepsis and Acute Kidney Injury Prediction." *44th IEEE International Conference on Computer Design (ICCD 2026)*, Hong Kong.

[C4] J. I. Thirupattur, S. V. Easwaramoorthy, A. Khalid, and **A. Rashid**. "MedFlow: A Drift-Adaptive Streaming Analytics Framework for Real-Time Medical Risk Prediction across Five Public Health Datasets." *44th IEEE International Conference on Computer Design (ICCD 2026)*, Hong Kong.

[C5] **R. A. R. Ejaz**†, F. Iradat, U. Taslim, Y. A. Bangash†, W. Iqbal, and M. B. K. Raja. "Nested Federated Learning: Layer-Wise Multi-Frequency Synchronisation for Privacy-Preserving Distributed Intelligence." *FAIEMA 2026* (Springer).

[C6] **R. A. R. Ejaz**†, F. Iradat, et al. "DP-f-FUM: Differentially Private Federated Unlearning via Per-Sample Clipping and Min–Max f-Divergence Optimisation." *FLTA 2026*, main track.

[C7] **R. A. R. Ejaz**, F. Iradat, and M. Kumail. "A Multi-Timescale Safety Architecture for Developmentally Adaptive Child–AI Interaction via Nested Policy Learning." *UK Workshop on Computational Intelligence (UKCI 2026)*, Coventry, UK.

[C8] **R. A. R. Ejaz**†, Q. M. Waiz, and F. Iradat. "HADES: Hierarchical Autonomous Digital Evidence System Using Skill-Driven Multi-Agent Orchestration for Digital Forensics." *Internationale Tagung Wirtschaftsinformatik (WI 2026)*, Linz, Austria.

[C9] **R. A. R. Ejaz** and F. Iradat. "Training-Free Coarse Point Cloud Registration via Axis-Projected Surface Area Matching." *ICCVDM 2026*.

[C10] A. M. Warris, Z. U. Ebad, **R. A. R. Ejaz**, and F. Iradat. "FedSRIMU: Spectral Roughness-Informed Federated Unlearning for Privacy-Compliant IoT Intrusion Detection." *UKCI 2026*, Coventry, UK.

[C11] A. Khan, A. Ahmedani, A. A. Kiani, F. Iradat, **R. A. R. Ejaz**, and Y. A. Bangash. "TEMPEST: Temporal Execution Modeling for Pre-Encryption Ransomware Detection with Timestep-Level Explainability." *ICSL-DSGA 2026*, Sunway University, Malaysia.

[C12] E. Tariq, E. Tariq, I. S. Khan, **R. A. R. Ejaz**, and F. Iradat. "From AI to Generative AI: How Emerging Technologies Are Reshaping Public Trust." *WI 2026*, Linz, Austria.

Under review

[R1] **R. A. R. Ejaz**, et al. "FrugalGuard: Cost-Aware Adaptive Escalation for Guarding Scoped Large Language Model Agents." *Array* (Elsevier).

[R2] **A. Rashid**, F. Iradat, W. Iqbal, F. A. Khan, and A. Ahmad. "From Assumption to Invariant: Supervisory Control for Byzantine-Robust Federated Learning on Mobile Devices." *IEEE Transactions on Mobile Computing*.

[R3] **A. Rashid**, F. Iradat, W. Iqbal†, B. Metin, and M. Mansouri. "LLM Intrusion Detection Agents Relocate the Privacy Exposure That Device-Layer Traffic Shaping Removes in Smart Home IoT Networks." *Discover Computing* (Springer).

[R4] **A. Rashid**, F. Iradat, and W. Iqbal†. "Molecular Prompt Injection: Hijacking Agentic Bio-Cyber Interfaces and a Nested Multi-Timescale Trust Monitor for the Internet of Bio-Nano Things." *Cognitive Computation* (Springer).

[R5] **R. A. R. Ejaz**, et al. "Descriptor-Space Transfer of Path-Loss Parameters in 6G Digital Twins: Degeneracy of Kernel Transfer Under Covariate Shift and Few-Shot Correction."

[R6] **R. A. R. Ejaz**, et al. "FedAdaPriv-CPU: Adaptive Differential Privacy with Frozen Backbone for Resource-Constrained Federated Medical Imaging." Springer Nature journal.

[R7] **R. A. R. Ejaz**, F. Iradat, W. Iqbal, and Y. A. Bangash. "NestAnt: A Nested-Learning Surrogate for Continual Multi-Band Microstrip Antenna Design." *FIT 2026*.

[R8] **R. A. R. Ejaz**, F. Iradat, et al. "Active-Ledger: Trust-Gated Diffusion and Behavioural Contribution Scoring for Adversarially Robust Federated ECG Arrhythmia Classification." *AJCAI 2026*.

[R9] **R. A. R. Ejaz**, F. Iradat, M. I. Kharka, and W. Iqbal. "Multi-Frequency Layer-Wise Aggregation for Federated Clinical Prediction Across Hospitals." *Int. J. of E-Health and Medical Communications* (IGI Global).

[R10] **R. A. R. Ejaz**, N. Aburaed, and F. Iradat. "Nested Learning for Multi-Timescale Electric Vehicle Charging Coordination: A Physics-Informed Deep RL Framework with Hard Voltage Guarantees."

HONOURS AND SELECTIONS

Best Paper Award, ICSL-DSGA 2026, Sunway University, Malaysia (with IIT Madras and the IEEE SMC Society), for [C1] — 2026

Selected participant, CISPA Helmholtz Center for Information Security Summer School, Federated Learning track — 2026

Selected participant, UK Quantum Hackathon (NQCC, University of Warwick): Rolls-Royce challenge on quantum methods for aerospace materials design — 2026

Selected to Pakistan's National AI Research Team at NCERT; Letter of Appreciation from its Director General — 2025

NESCOM research funding for undergraduate research on intelligent UAV communications — 2024

RESEARCH EXPERIENCE

AI Researcher and Co-Investigator, Telecommunications Research Lab (TRL), IBA Karachi — Jan 2025 – Present

Part-time, remote. With Dr. Faisal Iradat.

- Formulated layer-wise multi-frequency synchronisation for federated learning [C5], addressing client heterogeneity, communication cost and catastrophic forgetting in one framework; extended the idea to clinical prediction across hospitals [R9] and to continual antenna design [R7].

- Developed differentially private federated unlearning for right-to-erasure requests without full retraining [C6], and synthesised verified supervisory control for Byzantine-robust federated learning on mobile devices [R2].
- Lead author and coordinator of an international collaboration spanning the UK, US, UAE, Pakistan, Saudi Arabia and Oman; co-supervise the lab's undergraduate research projects (see Supervision).

AI Research Engineer, National Cyber Emergency Response Team (NCERT), Govt. of Pakistan Feb 2025 – Jun 2025
Islamabad, Pakistan

- Led the design and build of an LLM-based security information and event management system: Kafka ingestion, quantised inference, threat classification and automated alert routing.
- Automated Level-1 security operations centre triage across 200+ federal organisations at more than 25 million logs per day.

AI/ML Software Engineer, Electronic Media Services (EMS-UK): 6G digital twins and agentic AI Jul 2026 – Present
London, UK

- Build agentic-AI components for the IoDT2 CELTIC-NEXT 6G programme, linking simulation-grounded digital twins to LLM-driven orchestration with guardrails on tool use; this work led to [R5].
- Deployed a cloud-native digital-twin framework end to end on Kubernetes (Knative, RabbitMQ, PostgreSQL), with autoscaling, runbooks and operational handover.

SUPERVISION, TEACHING AND SERVICE

- **Co-supervisor of undergraduate research, IBA Karachi** (with Dr. Faisal Iradat), 2025 – present: federated unlearning for IoT intrusion detection [C10], multi-agent digital forensics [C8], pre-encryption ransomware detection [C11], and privacy in distributed information systems (Panopticon).
- **Industry trainer, IBA Centre for Information and Communication Technology:** designed and delivered an Agentic AI and Intelligent Automation programme for MOL Group, 2026.

PROFESSIONAL EXPERIENCE

Lead AI Transformation Architect (consultant), Global Exports, Pakistan 2026 – Present

- Recovered 823 business rules and 65 cross-module dependencies from a legacy Oracle ERP (125 Forms modules, 434 tables, 35,064 lines of PL/SQL), verified against the live schema with zero production impact; exposed the result to LLM agents through a read-only Model Context Protocol server.

AI/ML Engineer (contract), Curium: AI orchestration for ports and logistics Feb 2026 – Jun 2026

- Profiled and optimised the AI orchestration layer for port and logistics operations, reducing latency, throughput bottlenecks and compute cost.

AI Strategist and Technical Lead, AI Systems, Neura-X AI Consulting Sep 2023 – Jan 2025

- Architected a self-hosted multi-agent RAG platform with hybrid retrieval (dense, sparse, re-ranking), raising end-to-end answer accuracy from 76% to 89% at under 200 ms.
- Shipped ZazuAI, an on-premise meeting-intelligence system with real-time transcription under 500 ms (Whisper.cpp, DIART; 96% transcription and 80% diarisation accuracy after WhisperX post-processing).
- Built thermal-imaging models for industrial monitoring (ViT, CNN and spatio-temporal GNNs; 90 to 95% accuracy), reducing the ViT patch size from 16×16 to 4×4 for low-resolution input.

TECHNICAL SKILLS

Programming	Python (advanced), C++, SQL, PL/SQL, Bash, LaTeX
Machine learning	PyTorch, JAX, TensorFlow, Hugging Face, PyTorch Geometric, scikit-learn, XGBoost
LLMs and agents	LoRA/QLoRA fine-tuning, RAG, INT8/INT4 quantisation, agent orchestration, Model Context Protocol, evaluation and guardrails
Research practice	One-command reproducible pipelines, paired bootstrap tests, differential-privacy accounting, calibration
Systems	Kubernetes, Docker, Kafka, RabbitMQ, Redis, PostgreSQL, MLflow, DeepSpeed, FSDP, AWS
Languages	English (full professional proficiency), Urdu (native)

REFERENCES

Dr. Faisal Iradat,
Head, Telecommunications Research Lab,
IBA Karachi, Pakistan.
Research supervisor and mentor

Dr. Haider Abbas,
Director General, NCERT, Pakistan.
Employer reference (2025)

Dr. M. Waseem Iqbal,
Assistant Professor, Sultan Qaboos University, Oman.
Research collaborator

Contact details for referees are available on request.